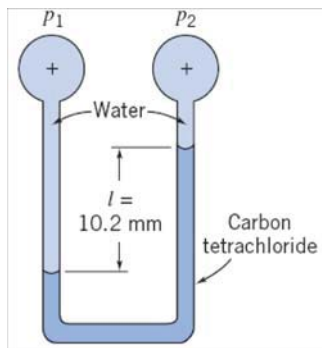
Problem 3.11

[Difficulty: 2]

3.11 Consider the two-fluid manometer shown. Calculate the applied pressure difference.



Given: Two-fluid manometer as shown

$l = 10.2 \cdot mm \quad SG_{ct} = 1.595$

Find: Pressure difference

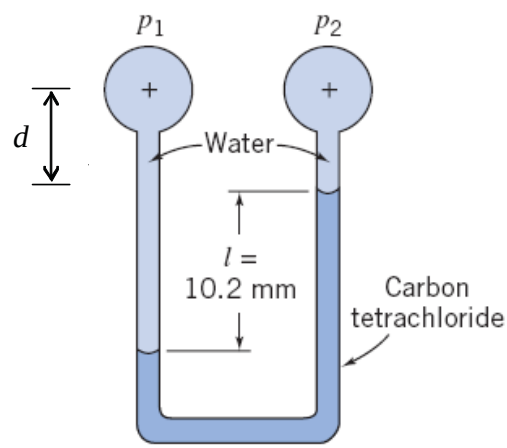
Solution: We will apply the hydrostatics equation.

Governing equations:

$$\frac{dp}{dh} = \rho \cdot g$$ (Hydrostatic Pressure - h is positive downwards)

$$\rho = SG \cdot \rho_{water}$$ (Definition of Specific Gravity)

Assumptions: (1) Static liquid
(2) Incompressible liquid



Starting at point 1 and progressing to point 2 we have:

$$p_1 + \rho_{water} \cdot g \cdot (d + l) - \rho_{ct} \cdot g \cdot l - \rho_{water} \cdot g \cdot d = p_2$$

Simplifying and solving for $p_2 - p_1$ we have:

$$\Delta p = p_2 - p_1 = \rho_{ct} \cdot g \cdot l - \rho_{water} \cdot g \cdot l = \left(SG_{ct} - 1\right) \cdot \rho_{water} \cdot g \cdot l$$

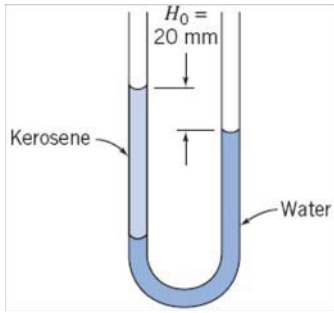
Substituting the known data:

$$\Delta p = (1.591 - 1) \times 1000 \cdot \frac{kg}{m^3} \times 9.81 \cdot \frac{m}{s^2} \times 10.2 \cdot mm \times \frac{m}{10^3 \cdot mm} \qquad \Delta p = 59.1\,Pa$$

Problem 3.12

[Difficulty: 2]

3.12 The manometer shown contains water and kerosene. With both tubes open to the atmosphere, the free-surface elevations differ by $H_0 = 20.0$ mm. Determine the elevation difference when a pressure of 98.0 Pa (gage) is applied to the right tube.



Given: Two fluid manometer contains water and kerosene. With both tubes open to atmosphere, the difference in free surface elevations is known

$$H_o = 20 \cdot mm \quad SG_k = 0.82$$

Find: The elevation difference, H, between the free surfaces of the fluids when a gage pressure of 98.0 Pa is applied to the right tube.

Solution: We will apply the hydrostatics equation.

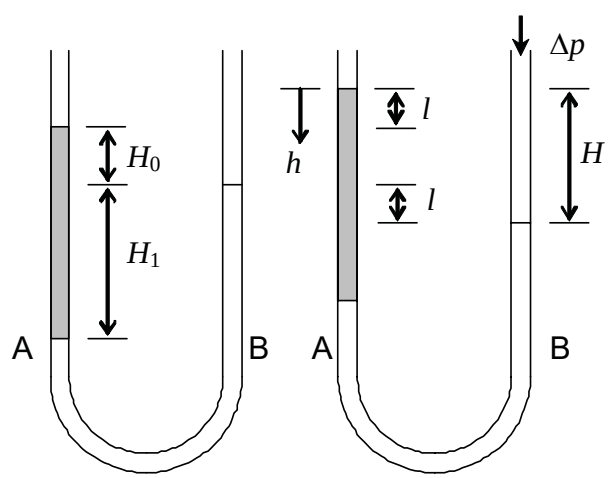
Governing Equations:

$$\frac{dp}{dh} = \rho \cdot g \qquad \text{(Hydrostatic Pressure - h is positive downwards)}$$

$$\rho = SG \cdot \rho_{water} \qquad \text{(Definition of Specific Gravity)}$$

Assumptions:
(1) Static liquid
(2) Incompressible liquid

When the gage pressure Δp is applied to the right tube, the water in the right tube is displaced downward by a distance, l. The kerosene in the left tube is displaced upward by the same distance, l.



Under the applied gage pressure Δp, the elevation difference, H, is:

$$H = H_o + 2 \cdot l$$

Since points A and B are at the same elevation in the same fluid, their pressures are the same. Initially:

$$p_A = \rho_k \cdot g \cdot (H_o + H_1) \qquad p_B = \rho_{water} \cdot g \cdot H_1$$

Setting these pressures equal:

$$\rho_k \cdot g \cdot (H_o + H_1) = \rho_{water} \cdot g \cdot H_1$$

Solving for H_1

$$H_1 = \frac{\rho_k \cdot H_o}{\rho_{water} - \rho_k} = \frac{SG_k \cdot H_o}{1 - SG_k} \qquad H_1 = \frac{0.82 \times 20 \cdot mm}{1 - 0.82} \qquad H_1 = 91.11 \cdot mm$$

Now under the applied gage pressure:

$$p_A = \rho_k \cdot g \cdot (H_o + H_1) + \rho_{water} \cdot g \cdot l \qquad p_B = \Delta p + \rho_{water} \cdot g \cdot (H_1 - l)$$

Setting these pressures equal:

$$SG_k \cdot \left(H_o + H_1\right) + l = \frac{\Delta p}{\rho_{water} \cdot g} + \left(H_1 - l\right) \qquad l = \frac{1}{2}\left[\frac{\Delta p}{\rho_{water} \cdot g} + H_1 - SG_k \cdot \left(H_o + H_1\right)\right]$$

Substituting in known values we get:

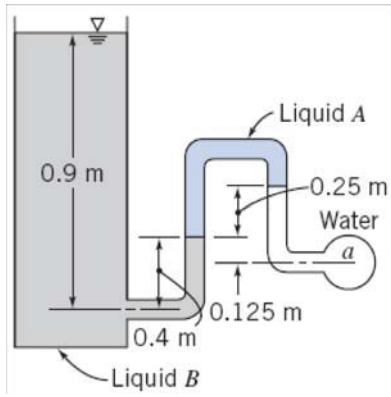
$$l = \frac{1}{2} \times \left[98.0 \cdot \frac{N}{m^2} \times \frac{1}{999}\frac{m^3}{kg} \times \frac{1}{9.81} \cdot \frac{s^2}{m} \times \frac{kg \cdot m}{N \cdot s^2} + [91.11 \cdot mm - 0.82 \times (20 \cdot mm + 91.11 \cdot mm)] \times \frac{m}{10^3 \cdot mm}\right] \qquad l = 5.000 \cdot mm$$

Now we solve for H:

$$H = 20 \cdot mm + 2 \times 5.000 \cdot mm \qquad H = 30.0 \cdot mm$$

Problem 3.13 [Difficulty: 2]

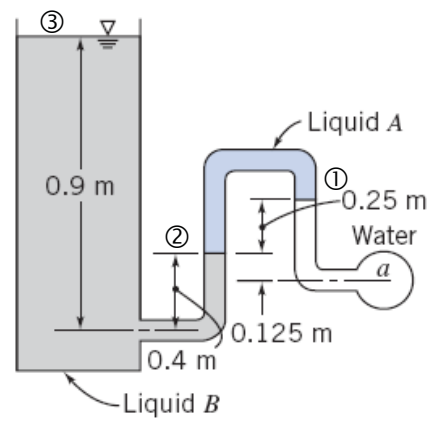
3.13 Determine the gage pressure in kPa at point a, if liquid A has SG = 1.20 and liquid B has SG = 0.75. The liquid surrounding point a is water, and the tank on the left is open to the atmosphere.



Given: Data on manometer

Find: Gage pressure at point a

Assumption: Water, liquids A and B are static and incompressible



Solution:

Basic equation $\frac{dp}{dy} = -\rho \cdot g$ or, for constant ρ $\Delta p = \rho \cdot g \cdot \Delta h$

where Δh is height difference

Starting at point a $p_1 = p_a - \rho_{H2O} \cdot g \cdot h_1$ where $h_1 = 0.125 \cdot m + 0.25 \cdot m$ $h_1 = 0.375\,m$

Next, in liquid A $p_2 = p_1 + SG_A \cdot \rho_{H2O} \cdot g \cdot h_2$ where $h_2 = 0.25 \cdot m$

Finally, in liquid B $p_{atm} = p_2 - SG_B \cdot \rho_{H2O} \cdot g \cdot h_3$ where $h_3 = 0.9 \cdot m - 0.4 \cdot m$ $h_3 = 0.5\,m$

Combining the three equations

$$p_{atm} = \left(p_1 + SG_A \cdot \rho_{H2O} \cdot g \cdot h_2\right) - SG_B \cdot \rho_{H2O} \cdot g \cdot h_3 = p_a - \rho_{H2O} \cdot g \cdot h_1 + SG_A \cdot \rho_{H2O} \cdot g \cdot h_2 - SG_B \cdot \rho_{H2O} \cdot g \cdot h_3$$

$$p_a = p_{atm} + \rho_{H2O} \cdot g \cdot \left(h_1 - SG_A \cdot h_2 + SG_B \cdot h_3\right)$$

or in gage pressures $p_a = \rho_{H2O} \cdot g \cdot \left(h_1 - SG_A \cdot h_2 + SG_B \cdot h_3\right)$

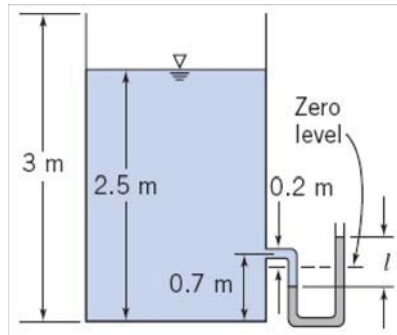
$$p_a = 1000 \cdot \frac{kg}{m^3} \times 9.81 \cdot \frac{m}{s^2} \times [0.375 - (1.20 \times 0.25) + (0.75 \times 0.5)] \cdot m \times \frac{N \cdot s^2}{kg \cdot m}$$

$p_a = 4.41 \times 10^3\,Pa$ $p_a = 4.41 \cdot kPa$ (gage)

Problem 3.16

[Difficulty: 2]

3.16 A rectangular tank that is open to the atmosphere is filled to a depth of 2.5 m. A U-tube manometer filled with Meriam blue manometer fluid (SG = 1.75) is connected to the tank 0.7 m above the tank bottom. Before the manometer is connected to the tank, the zero level is 0.2 m below the connection. Determine the deflection after the manometer is connected.



Given: A U-tube manometer is connected to the open tank filled with water as shown (manometer fluid is Meriam blue)

$D_1 = 2.5 \cdot m \quad D_2 = 0.7 \cdot m \quad d = 0.2 \cdot m \quad SG_{oil} = 1.75$

Find: The manometer deflection, l

Solution: We will apply the hydrostatics equations to this system.

Governing Equations:

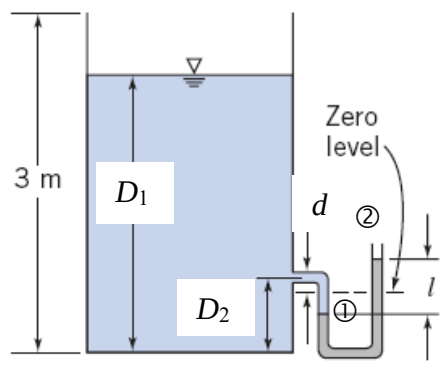
$$\frac{dp}{dh} = \rho \cdot g \qquad \text{(Hydrostatic Pressure - h is positive downwards)}$$

$$\rho = SG \cdot \rho_{water} \qquad \text{(Definition of Specific Gravity)}$$

Assumptions: (1) Static liquid
(2) Incompressible liquid

Integrating the hydrostatic pressure equation we get:

$$\Delta p = \rho \cdot g \cdot \Delta h$$



When the tank is filled with water, the oil in the left leg of the manometer is displaced downward by l/2. The oil in the right leg is displaced upward by the same distance, l/2.

Beginning at the free surface of the tank, and accounting for the changes in pressure with elevation:

$$p_{atm} + \rho_{water} \cdot g \cdot \left(D_1 - D_2 + d + \frac{l}{2}\right) - \rho_{oil} \cdot g \cdot l = p_{atm}$$

Upon simplification:

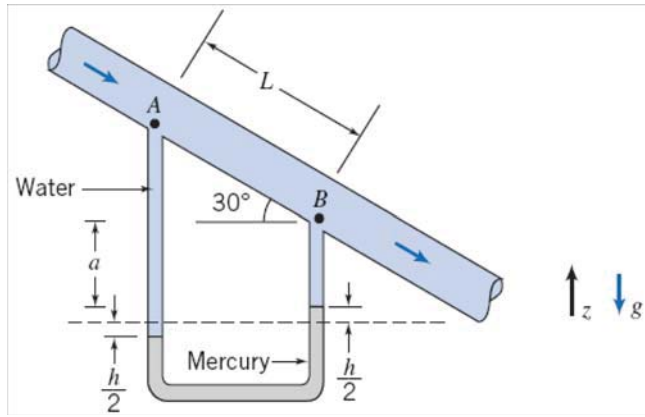
$$\rho_{water} \cdot g \cdot \left(D_1 - D_2 + d + \frac{l}{2}\right) = \rho_{oil} \cdot g \cdot l \qquad D_1 - D_2 + d + \frac{l}{2} = SG_{oil} \cdot l \qquad l = \frac{D_1 - D_2 + d}{SG_{oil} - \frac{1}{2}}$$

$$l = \frac{2.5 \cdot m - 0.7 \cdot m + 0.2 \cdot m}{1.75 - \frac{1}{2}} \qquad l = 1.600\,m$$

Problem 3.19

[Difficulty: 2]

3.19 Water flows downward along a pipe that is inclined at 30° below the horizontal, as shown. Pressure difference $p_A - p_B$ is due partly to gravity and partly to friction. Derive an algebraic expression for the pressure difference. Evaluate the pressure difference if $L = 5$ ft and $h = 6$ in.



Given: Water flow in an inclined pipe as shown. The pressure difference is measured with a two-fluid manometer

$L = 5 \cdot ft \qquad h = 6 \cdot in \qquad SG_{Hg} = 13.55$

Find: Pressure difference between A and B

Solution: We will apply the hydrostatics equations to this system.

Governing Equations:

$$\frac{dp}{dh} = \rho \cdot g \qquad \text{(Hydrostatic Pressure - h is positive downwards)}$$

$$\rho = SG \cdot \rho_{water} \qquad \text{(Definition of Specific Gravity)}$$

Assumptions:
(1) Static liquid
(2) Incompressible liquid
(3) Gravity is constant

Integrating the hydrostatic pressure equation we get:

$$\Delta p = \rho \cdot g \cdot \Delta h$$

Progressing through the manometer from A to B:

$$p_A + \rho_{water} \cdot g \cdot L \cdot \sin(30 \cdot deg) + \rho_{water} \cdot g \cdot a + \rho_{water} \cdot g \cdot h - \rho_{Hg} \cdot g \cdot h - \rho_{water} \cdot g \cdot a = p_B$$

Simplifying terms and solving for the pressure difference:

$$\Delta p = p_A - p_B = \rho_{water} \cdot g \cdot \left[h \cdot \left(SG_{Hg} - 1 \right) - L \cdot \sin(30 \cdot deg) \right]$$

Substituting in values:

$$\Delta p = 1.94 \cdot \frac{slug}{ft^3} \times 32.2 \frac{ft}{s^2} \times \left[6 \cdot in \times \frac{ft}{12 \cdot in} \times (13.55 - 1) - 5 \cdot ft \times \sin(30 \cdot deg) \right] \times \frac{lbf \cdot s^2}{slug\, ft} \times \left(\frac{ft}{12 \cdot in} \right)^2 \qquad \Delta p = 1.638 \cdot psi$$

Problem 8.51 [Difficulty: 2]

8.51 A 75-mm-diameter orifice with D and D/2 taps is used to measure the flow rate of 65°Cwater in a 150-mm-ID pipe. Determine the pressure difference for a flow rate of 20 L/s.

Given: Flow through an orifice

Find: Pressure drop

From Dr. Latulippe: In all the equations below simply replace K with Co in order to match the notation in the lecture notes and Figure 8.20

Solution:

Basic equation $m_{actual} = K \cdot A_t \cdot \sqrt{2 \cdot \rho \cdot (p_1 - p_2)} = K \cdot A_t \cdot \sqrt{2 \cdot \rho \cdot \Delta p}$

Note that m_{actual} is mass flow rate (the software cannot render a dot!)

For the flow coefficient $K = K\left(Re_{D1}, \frac{D_t}{D_1}\right)$

At 65°C,(Table A.8) $\rho = 980 \cdot \frac{kg}{m^3}$ $\nu = 4.40 \times 10^{-7} \cdot \frac{m^2}{s}$

$V = \frac{Q}{A}$ $V = \frac{4}{\pi} \times \frac{1}{(0.15 \cdot m)^2} \times 20 \cdot \frac{L}{s} \times \frac{0.001 \cdot m^3}{1 \cdot L}$ $V = 1.13 \frac{m}{s}$

$Re_{D1} = \frac{V \cdot D}{\nu}$ $Re_{D1} = 1.13 \cdot \frac{m}{s} \times 0.15 \cdot m \times \frac{s}{4.40 \times 10^{-7} \cdot m^2}$ $Re_{D1} = 3.85 \times 10^5$

$\beta = \frac{D_t}{D_1}$ $\beta = \frac{75}{150}$ $\beta = 0.5$

From Fig. 8.20 $K = 0.624$

Then $\Delta p = \left(\frac{m_{actual}}{K \cdot A_t}\right)^2 \cdot \frac{1}{2 \cdot \rho} = \left(\frac{\rho \cdot Q}{K \cdot A_t}\right)^2 \cdot \frac{1}{2 \cdot \rho} = \frac{\rho}{2} \cdot \left(\frac{Q}{K \cdot A_t}\right)^2$

$$\Delta p = \frac{1}{2} \times 980 \cdot \frac{kg}{m^3} \times \left[20 \cdot \frac{L}{s} \times \frac{0.001 \cdot m^3}{1 \cdot L} \times \frac{1}{0.624} \times \frac{4}{\pi} \times \frac{1}{(0.075 \cdot m)^2}\right]^2 \qquad \Delta p = 25.8 \cdot kPa$$